

## Chapter 11

### Introduction(S.H.M)

#### VIBRATORY/OSCILLATORY MOTION

Vibratory/oscillatory motion is a type of periodic motion in which an object moves back and forth about a fixed point. Oscillatory motion is a periodic motion which repeats itself after an equal period of time.

Some examples of vibratory motion:

- Sound waves:
- Pendulum:
- Tuning fork: A tuning fork is a metal fork with two prongs that vibrate when struck.

#### BASIC POINTS TO UNDERSTAND VIBRATORY MOTION

|              |                                                                                                                                               |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| VIBRATION    | The travelling of a vibratory body from one extreme position to the opposite and back again is called one vibration.                          |
| TIME PERIOD  | It is the time required by a vibratory body to complete one vibration. It is denoted by (T)                                                   |
| FREQUENCY    | It is number of vibrations completed in one second. It is denoted by f. Its units are cycle/ Sec or hertz or vibration / Sec.                 |
| DISPLACEMENT | It is the distance of a vibrating body from the equilibrium or mean position to the position of the body at any instant. It is denoted by (x) |
| AMPLITUDE    | It is the maximum displacement of a vibrating body on either side of its equilibrium position. It is denoted by ( $x_0$ )                     |
| PHASE        | It is defined as ratio between instantaneous displacement (x) and amplitude ( $x_0$ ).                                                        |

#### SIMPLE HARMONIC MOTION (S.H.M)

It is a type of vibrating motion in which the acceleration is directly proportional to the displacement of body and is always directed to the mean or equilibrium position.

$$a \propto -x$$

Where negative sign shows that the acceleration (a) is oppositely directed to the displacement (x)

#### Condition / Characteristics Of S.H.M:

- The body should exhibit to and fro vibrating motion.
- The acceleration of the body at any instant should be directly proportion to its displacement.  
 $a \propto x$
- The acceleration should be directed towards the mean position indicated by the negative sign  
 $a \propto -x$

- The body must possess inertia.
- The body must be acted upon by a restoring force

#### Example of S.H.M:

- The to and fro motion of a body attached to a horizontal mounted spring.
- The vibratory motion of a body attached to a vertically suspended spring.
- A Simple Pendulum.
- A swing.
- Projection of a point mass moving with uniform angular velocity



### Motion Of An Oscillator (Experimental Methods)

Let an object of mass ( $m$ ) is attached to a spring as shown in a figure.

Let a force is applied, due to which the body is displacement by a distance  $x$ .

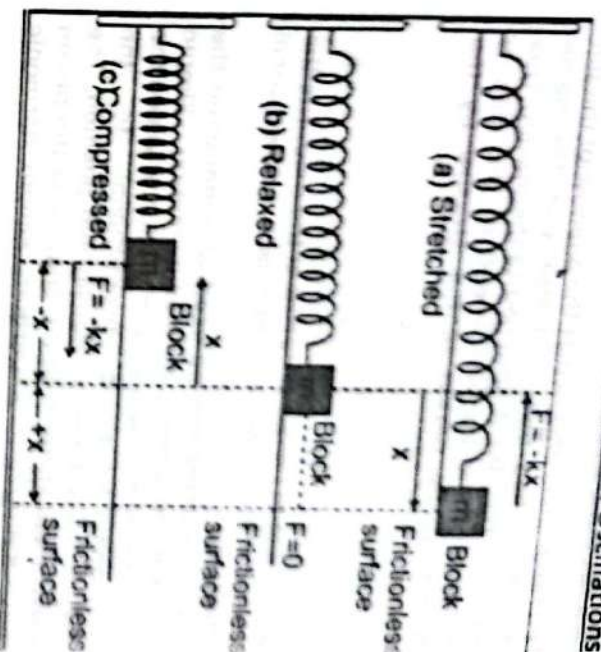
CH#11: Oscillations

According to Hooke's Law

"Applied force is directly proportional to the displacement."

$$F \propto x$$

$$F = kx$$



Where  $k$  is spring constant.

Since the elastic restoring force ( $F$ ) is opposite to the applied force, therefore

$$F = -kx \quad (1)$$

If " $a$ " is the acceleration of body at " $x$ " displacement. According to the Newton's second law of motion.

$$F = ma \quad (2)$$

Comparing (1) and (2)

$$ma = -Kx$$

$$a = -\frac{K}{m}x$$

$$a = -(\text{constant})x$$

Since  $k/m$  is constant  
Or

$$a \propto -x$$

This is equation for S.H.M.

This shows that "acceleration ( $a$ ) of the body is directly proportional to its displacement and is directed to mean position".

Hence the motion of body under elastic restoring motion force is S.H.M.

#### Graphical Representation Of S.H.M

- We attach an object to a spring and let it oscillate. The maximum displacement of the object from its equilibrium position is  $x_0$  (amplitude).
- At the same time, we place a pin on a horizontal circular disc of radius  $R = x_0$ . The disc is set into motion with constant angular speed  $\omega$ .
- The object and the disc are set into motion at the same time.
- We shine a light on the pin and the object so that their shadows are projected onto the screen.
- We adjust the speed of the disc until the shadows of the pin and the object oscillate with the same period.

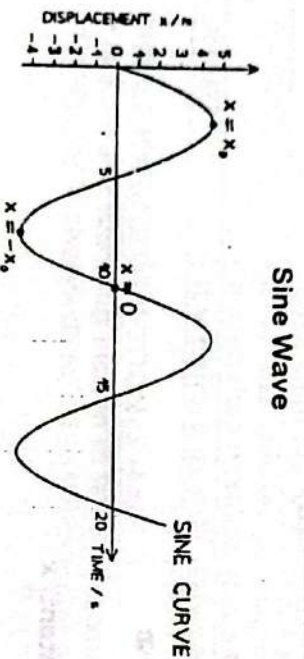
The experiment shows that the shadow of the pin on the rotating disc oscillates with the same period as the object attached to the spring. This is because the angular speed of the disc is related to the period of the shadow by the equation:

$$\omega = \frac{2\pi}{T}$$

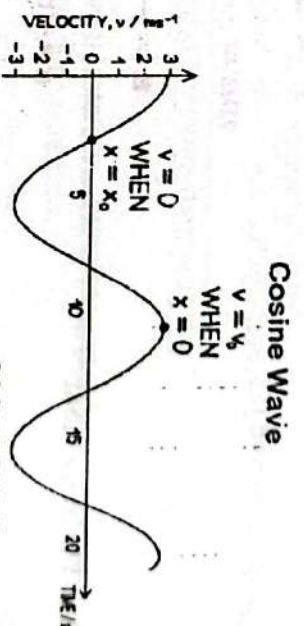
where  $\omega$  is the angular speed of the disc,  $T$  is the period of the shadow, and  $\pi$  is a mathematical constant.

The experiment can be used to understand the concept of simple harmonic motion. Simple harmonic motion is a type of motion in which the restoring force is proportional to the displacement from the equilibrium position. The experiment shows that the shadow of the pin on the rotating disc undergoes simple harmonic motion

Displacement Time Graph:



$$x(t) = x_0 \sin \theta = x_0 \sin \omega t$$



$$x(t) = x_0 \cos \theta = x_0 \cos \omega t$$

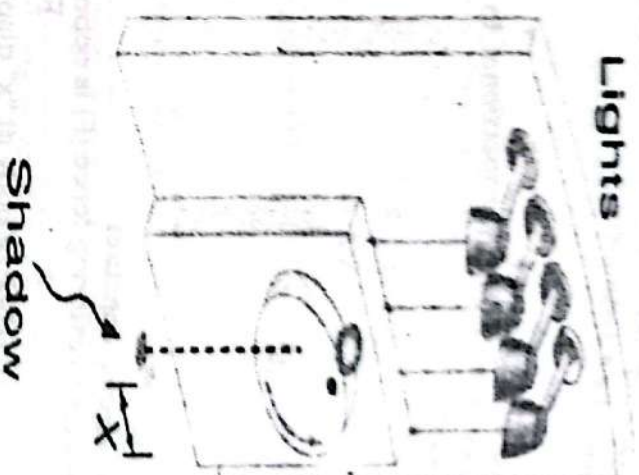
## Chapter 11

### Projection of particle moving in a circular path

#### MOTION OF THE PROJECTION OF A PARTICLE MOVING ALONG A CIRCULAR PATH:

Let a Particle "P" is moving in a circle of radius " $r$ " with constant linear Velocity (" $V_p$ ") and angular Velocity ( $\omega$ ) so that

$$V_p = r\omega \quad \text{--- (1)}$$





Let  $Q$  be the projection of  $P$  and  $x$  be the displacement from mean point  $O$  at any instant.

Let  $a_c$  = Centripetal acceleration of  $P$   
 $a$  = Linear acceleration of  $Q$

**Centripetal Acceleration:**  
 Centripetal acceleration of  $P$  is given by

$$a_c = \frac{v_p^2}{r}$$

But  $v_p = r\omega$

$$a_c = \omega^2 r^2 / r$$

$$a_c = \omega^2 r \quad \text{where } r = x_0$$

$$\text{Or } a_c = \omega^2 x_0 \quad \text{---(2)}$$

Now consider  $\triangle OPQ$  for acceleration " $a$ "

Where

$$\cos \theta = \frac{\text{Base}}{\text{Hypoteneous}} = \frac{OQ}{OP}$$

$$\cos \theta = \frac{a}{a_c}$$

$$a = a_c \cos \theta$$

Putting value of  $a_c = \omega^2 r$  or  $a_c = \omega^2 x_0$   
 We have

$$a = \omega^2 x_0 \cos \theta \quad \text{---(3)}$$

Now again consider  $\triangle OPQ$

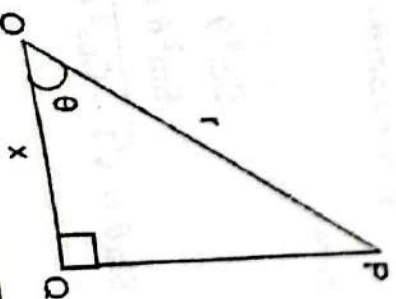
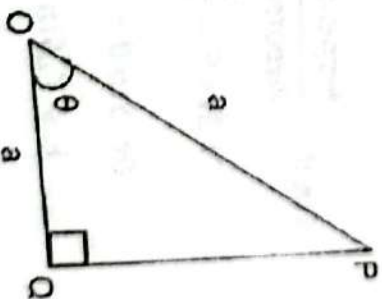
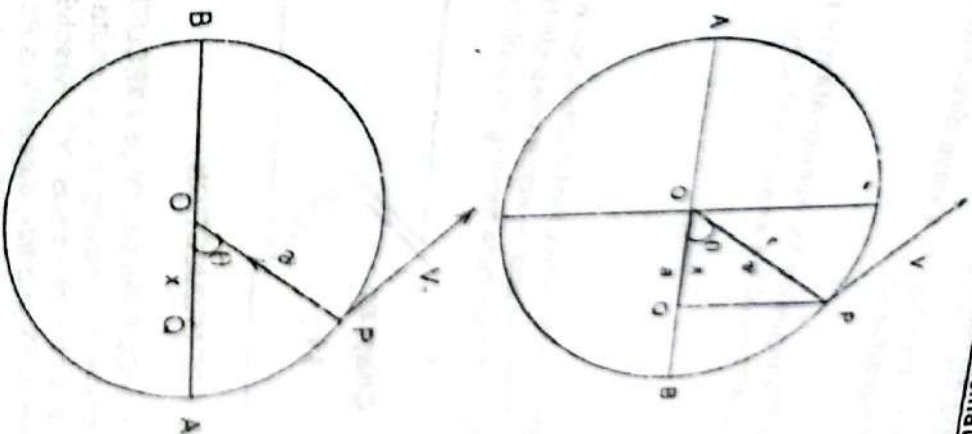
$$\cos \theta = \frac{\text{Base}}{\text{Hypoteneous}} = \frac{OQ}{OP}$$

$$\cos \theta = \frac{x}{r} = \frac{x}{x_0}$$

Putting this value of  $\cos \theta$  in Equation (3)  
 We have

$$a = \omega^2 x_0 (x / x_0)$$

$$a = \omega^2 x$$



Since acceleration is always directed to the mean position, therefore we put a negative sign in the above equation i.e.

$$a = -\omega^2 x$$

This is the equation for acceleration of projection of particle P moving in circle. As particle P is moving with constant angular velocity ( $\omega$ )

$$a = -(\text{Constant}) x$$

Or

$$a \propto -x$$

This expression shows that acceleration of projection (Q) of a particle P is directly proportional to its displacement (x) and directed towards the mean position (O). Hence we can conclude that the projection of a particle moving on a circumference of a circle of radius "r" executes S.H.M.

## Chapter 11

## VELOCITY OF PROJECTION

### VELOCITY OF PROJECTION

#### INSTANTANEOUS VELOCITY OF EXECUTING S.H.M:

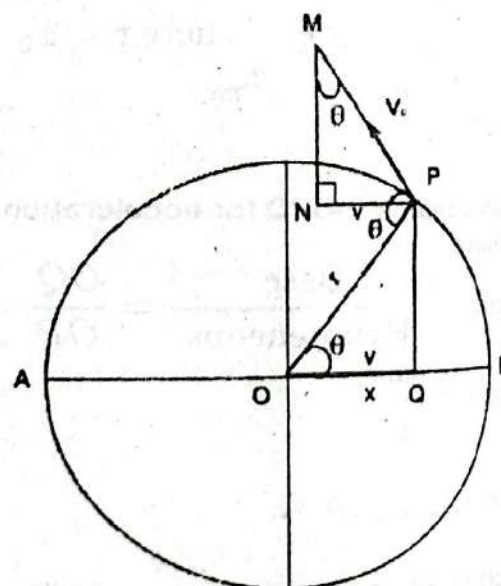
Let a particle P is moving in the constant linear speed ( $V_p$ ) in a circle of radius "r". Resolving  $V_p$  into its components PN and MN. Since PN is Parallel to OQ, so PN represents the velocity V of projection Q.

$$\sin \theta = \frac{\text{Perpendicular}}{\text{Hypoteneous}}$$

$$\text{Or } \sin \theta = \frac{PN}{PM}$$

$$\text{Or } \sin \theta = \frac{V}{V_p}$$

$$V = V_p \sin \theta$$



But  $V_p = r\omega$

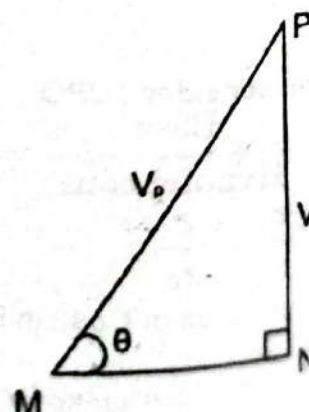
$$V = r\omega \sin \theta \dots (1)$$

Now

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin \theta = \sqrt{1 - \cos^2 \theta} \dots (2)$$



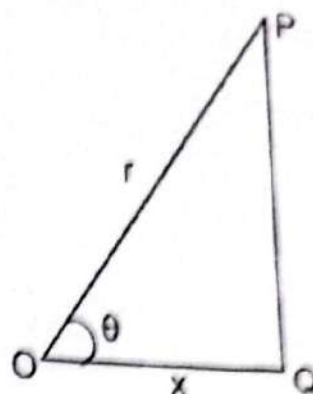


$$\cos \theta = \frac{\text{Base}}{\text{Hypotenuse}} = \frac{OQ}{OP} = \frac{x}{r}$$

Substitute the value of  $\cos \theta$  in equation (2)

$$\sin \theta = \sqrt{1 - (x/r)^2}$$

$$\sin \theta = \sqrt{\frac{r^2 - x^2}{r^2}}$$



By substituting the value of  $\sin \theta$  in equation (1) we have

$$V = r\omega \sqrt{\frac{r^2 - x^2}{r^2}}$$

Or

$$V = \omega \sqrt{r^2 - x^2}$$

Or

$$V = \omega \sqrt{x_0^2 - x^2}$$

[ $\because r = x_0 = \text{Maximum displacement or amplitude}$ ]

$$\left[ \text{as we know } \omega = \sqrt{\frac{k}{m}} \right]$$

$$V_{\text{ins}} = \sqrt{\frac{k}{m}} \sqrt{x_0^2 - x^2}$$

This is the expression for instantaneous velocity

**Maximum Velocity Of A Body Executing S.H.M:**

The projection Q has maximum velocity when its mean position "O", its displacement is zero ( $x=0$ ).  
By putting the value of  $x$  into the equation of Instantaneous velocity.

$$V_{\text{max}} = \sqrt{\frac{k}{m}} \sqrt{x_0^2}$$

$$V_{\text{max}} = \sqrt{\frac{k}{m}} x_0$$

### Relationship Between Maximum And Instantaneous Velocity:

As we know that

$$V_{\text{ins}} = \sqrt{\frac{k}{m}} \sqrt{x_0^2 - x^2}$$

$$V_{\text{ins}} = \sqrt{\frac{k}{m}} x_0 \sqrt{1 - \frac{x^2}{x_0^2}}$$

$$V_{\text{ins}} = V_{\text{max}} \sqrt{1 - \frac{x^2}{x_0^2}}$$

### Important Term Used In S.H.M

- **Instantaneous displacement (x)** is the distance of the oscillating particle from its equilibrium position at any instant. For example, if the oscillating particle is a mass attached to a spring, the instantaneous displacement is the distance between the mass and the spring's equilibrium position.
- **Amplitude ( $x_0$ )** is the maximum displacement of the oscillating particle from its equilibrium position. It is the distance from the equilibrium position to the highest or lowest point (extreme positions) of the oscillation.
- **Time period (T)** is the time it takes for the oscillating particle to complete one full oscillation. For example, if the oscillating particle is a mass attached to a spring, the time period is the time it takes for the mass to travel from its equilibrium position to the farthest point away from the equilibrium position and back again.
- **Frequency (f)** is the number of oscillations per unit time. It is the reciprocal of the time period. For example, if the oscillating particle is a mass attached to a spring, the frequency is the number of times the mass passes through its equilibrium position per second.
- **Angular frequency ( $\omega$ )** is the rate of change of the angular displacement of the oscillating particle. It is the number of radians the particle rotates per unit time. The angular frequency is related to the frequency by the equation  $\omega = 2\pi f$ .
- **Phase ( $\phi$ )** is the angular displacement of the oscillating particle at a given time. It is measured in radians. The phase can be used to track the progress of an oscillation.  
Or  
It is defined as ratio between instantaneous displacement (x) and amplitude ( $x_0$ ).

## Chapter 11

## Motion of a Mass Attached to a Vertical Spring

Some of the practical examples of S.H.M are

- Spring Mass System
- Simple Pendulum

### MOTION OF A MASS ATTACHED TO A VERTICAL SPRING:

Figure shows a coil spring of force constant "k" and no-load of length "L". When a body of mass "m" is attached to the spring it hangs in equilibrium with the spring extended by an amount " $\Delta L$ ", such that the restoring or upward force exerted by the spring is equal to the weight of the body of mass "m".



But from the Hook's Law

$$F \propto -\Delta L$$

$$F = -k\Delta L$$

Ignoring negative sign as it only represent restoration.

$$F = k\Delta L \quad \text{-----(1)}$$

Also

$$F_{\text{res.}} = W = mg \quad \text{-----(2)}$$

By comparing both equations:

$$k\Delta L = mg \quad \text{-----(3)}$$

We can write

$$F_{\text{Net}} = k\Delta L - mg \quad \text{-----(4)}$$

Now suppose the body of mass "m" is at distance "y" above its equilibrium position as in figure C. The extension of the spring is now  $(\Delta L - y)$ . Now the restoring force is  $k(\Delta L - y)$ . so eq (1) can be written as :

$$F = k(\Delta L - y)$$

i.e

The net force in this condition is

$$F_{\text{Net}} = k(\Delta L - y) - mg = k\Delta L - ky - mg \quad \text{from eq (3)}$$

$$k\Delta L = mg$$

$$F_{\text{Net}} = mg - ky - mg$$

$$F_{\text{Net}} = -ky$$

Since "k" is a constant , therefore:

$$F_{\text{Net}} \propto -y$$

It shows vertical mass system exhibit S.H.M.

## Chapter 11

## SIMPLE PENDULUM

### SIMPLE PENDULUM

#### Definition

A simple pendulum consists of mass "m" suspended by a light flexible and inextensible string.



## Physics XI

## Basic Assumptions

1. The radius of bob is supposed to be small as compare to the length of thread.
2. String is weightless and inextensible.
3. Support is frictionless.
4. Amplitude is supposed to be small i.e. angle of oscillation is so small.

**Prove Vibratory Motion Of Pendulum is S.H.M**  
 Consider a pendulum having a bob of mass "m" suspended by a string of length "L". Let x be the instantaneous displacement of bob from mean position, when it is vibrating.

The forces exerting on the bob are  
 (i) Weight "W" of bob acting down.  
 (ii) Tension "T" in the string acting upward

Resolving W into its components

- (a)  $W_y = W \sin \theta = mg \sin \theta$   
(Perpendicular to string)
- (b)  $W_x = W \cos \theta = mg \cos \theta$   
(Along the string)

Since the bob has no motion along the string,  
 so

$$T = W_x$$

$$T = W \cos \theta \quad \text{--- (1)}$$

This is expression for Tension (T) in the string.

Now the bob is vibrating due to force  $W_y$  or  $mg \sin \theta$ . Hence resorting force (F) on the bob is

$$F = -mg \sin \theta$$

$$ma = -mg \sin \theta$$

$$a = -g \sin \theta \quad \text{--- (2)}$$

The Pendulum is given a small displacement, hence  $\theta$  is very small; so

$$\sin \theta = \theta \text{ (in radian)}$$

$$a = -g\theta \quad \text{--- (3)}$$

At this stage we can conclude that acceleration is directly proportional to angle of oscillation negative sign shows that the direction of acceleration is opposite to the direction of angle of oscillation

Consider triangle  $\Delta OAB$

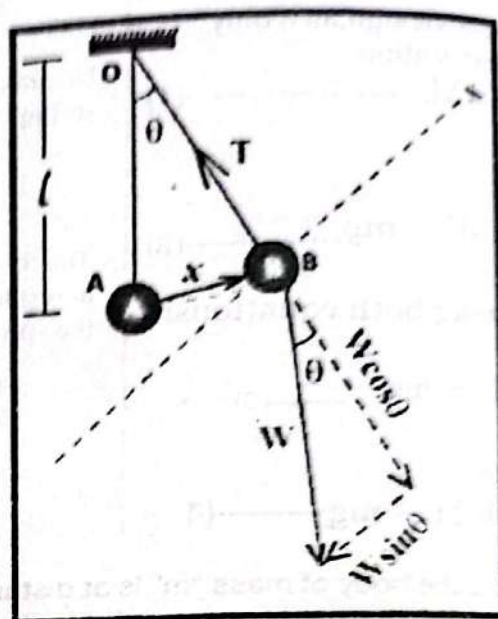
$$\sin \theta = \frac{\text{Perpendicular}}{\text{Hypotenous}}$$

$$\sin \theta = \frac{AB}{OA}$$

$$\sin \theta = \frac{x}{l}$$

$$\theta = \frac{x}{l}$$

[ as  $\theta$  is very small therefore  $\sin \theta \approx \theta$  ]



By substituting the value of (3) into equation (3)

$$a = -\frac{x}{l} g$$

$$a = -\left(\frac{g}{l}\right)x \dots \dots (4)$$

This is the expression for the acceleration of simple pendulum.  
Now for a given pendulum  $g/l$  is constant, so

$$a = -(\text{Constant})x$$

$$a \propto -x$$

This is equation for SHM, which shows that acceleration of pendulum is directly proportional to displacement and directed to its mean position. Hence motion of pendulum is SHM.

**Time Period Of Simple Pendulum:**

In S.H.M the acceleration of the vibrating body is expressed as

$$a = -\omega^2 x$$

But for simple pendulum

$$a = -\left(\frac{g}{l}\right)x$$

Comparing both above equations we get.

$$-\omega^2 x = -\frac{g}{l}x$$

$$\omega^2 = \frac{gx}{lx}$$

$$\omega^2 = \frac{g}{l}$$

$$\omega = \sqrt{\frac{g}{l}}$$

As we know that

$$T = \frac{2\pi}{\omega}$$

Put the value of  $\omega$  in the above equation we get.

$$T = 2\pi\sqrt{\frac{l}{g}}$$

**Frequency Of Simple Pendulum:**

As we know that, the frequency is the reciprocal of time period, i.e.



$$f = \frac{1}{T}$$

$$f = \frac{1}{2\pi\sqrt{l/g}}$$

$$f = \frac{1}{2\pi}\sqrt{\frac{g}{l}}$$

**LAW OF SIMPLE PENDULUM:**

- Time Period of simple pendulum is directly proportional to the square root of its length if "g" is constant.

$$T \propto \sqrt{L}$$

- Time period of simple pendulum is inversely proportional to the square root of acceleration due to gravity.

$$T \propto \frac{1}{\sqrt{g}}$$

- Time period of a simple pendulum is independent of amplitude if all other conditions kept constant.
- Time period of a simple pendulum is independent of mass of pendulum if all other conditions kept constant.

## Chapter 11

## CONSERVATION OF ENERGY

**LAW OF CONSERVATION OF ENERGY IN OSCILLATION:**

In oscillatory motion the law of conservation of energy can be verified as follow:

**Instantaneous Kinetic Energy Of A Body Executing S.H.M:**

As we know that by the definition of kinetic energy

$$K = \frac{1}{2}mv^2$$

When a body executing S.H.M

$$K_{ins} = \frac{1}{2}mv_{ins}^2$$

By putting the value of  $V_{ins}$  in above equation.

$$\text{Where, } V_{ins} = \sqrt{\frac{k}{m}} \sqrt{x_0^2 - x^2}$$

$$K_{\text{ins}} = \frac{1}{2} m \left( \sqrt{\frac{k}{m} \sqrt{x_0^2 - x^2}} \right)^2$$

$$K_{\text{ins}} = \frac{1}{2} m \left( \frac{k}{m} \right) (x_0^2 - x^2)$$

$$K_{\text{ins}} = \frac{1}{2} k (x_0^2 - x^2)$$

### Maximum Kinetic Energy Of A Body Executing S.H.M:

As we know that by the definition of kinetic energy

$$K = \frac{1}{2} mv^2$$

When a body executing S.H.M at mean position

$$K_{\text{max}} = \frac{1}{2} mv_{\text{max}}^2$$

By putting the value of  $v_{\text{max}}$  in above equation.

Where as

At mean position  $x = 0$

$$v_{\text{max}} = \sqrt{\frac{k}{m}} x_0 \text{ and } v_{\text{max}}^2 = \left( \frac{k}{m} \right) x_0^2$$

$$K_{\text{max}} = \frac{1}{2} k x_0^2$$

### Instantaneous Potential Energy Of A Body Executing S.H.M:

Potential energy (U) of a body executing S.H.M. is equal to the work done in displacing it from one point to another.

$$U = \text{WORK} = \text{FORCE} \times \text{DISPLACEMENT}$$

But restoring force at projection of P is not constant that is why we take the average of two forces at mean position and at any instantaneous position.

At any instance when displacement is " $x$ ". The restoring force will be.

$$F_1 = kx$$

At mean position where displacement is zero ( $x = 0$ ) restoring force will be

$$F_2 = 0$$

Hence the average restoring force will be.



$$F = \frac{F_1 + F_2}{2}$$

$$F = \frac{kx + 0}{2}$$

$$F = \frac{kx}{2}$$

since

$$U_{\text{ins}} = \text{Work} = F \times [\text{Displacement}]$$

$$U_{\text{ins}} = \frac{kx}{2} (x)$$

$$U_{\text{ins}} = \frac{1}{2} k x^2$$

### Maximum Potential Energy Of A Body Executing S.H.M:

P.E is maximum when displacement is maximum i.e. equal to amplitude ( $x_0$ ), when projection is extreme position A or B.

i.e: when  $x = x_0$  and  $U_{\text{ins}} = U_{\text{max}}$

$$U_{\text{max}} = \frac{1}{2} k x_0^2$$

### Total Energy Of A Body Executing S.H.M:

- At extreme points A OR B where displacement is maximum .

$$U_{\text{max}} = \frac{1}{2} k x_0^2$$

$$K_{\text{min}} = 0$$

$$\text{Total Energy} = U_{\text{max}} + K_{\text{min}} = \frac{1}{2} k x_0^2$$

- At mean position "O" where velocity is maximum and displacement is zero.

$$U_{\text{min}} = 0$$

$$K_{\text{max}} = \frac{1}{2} k x_0^2$$

$$\text{Total Energy} = U_{\text{min}} + K_{\text{max}} = \frac{1}{2} k x_0^2$$

- At all other points, where displacements is  $x$

$$K_{\text{ins}} = \frac{1}{2} k (x_0^2 - x^2)$$

$$U_{\text{ins}} = \frac{1}{2} k x^2$$

$$\text{Total Energy} = K_{\text{ins}} + U_{\text{ins}} = \frac{1}{2} k (x_0^2 - x^2) + \frac{1}{2} k x^2 = \frac{1}{2} k x_0^2$$

From eq(1), eq(2) and eq(3) we can say that Total Energy will remain constant.



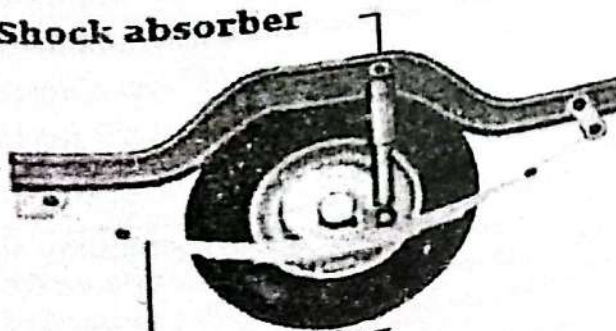
- Free oscillation** is an oscillation that occurs when a system is displaced from its equilibrium position and then released. The system will oscillate back and forth with a constant amplitude and frequency. An example of a free oscillation is the motion of a pendulum.
- Forced oscillation** is an oscillation that occurs when a system is subjected to a periodic force. The frequency of the forced oscillation will be equal to the frequency of the external force. An example of a forced oscillation is the vibration of a guitar string when it is plucked.
- Damped oscillation** is an oscillation that loses energy over time due to friction or other resistive forces. The amplitude of the oscillation will decrease gradually over time, and the swing that is not being pushed. An example of a damped oscillation is the motion of a pendulum that is not being pushed.
- Resonance** is a phenomenon that occurs when the frequency of a forced oscillation is equal to the natural frequency of the system. In this case, the amplitude of the oscillation will be greatly increased. An example of resonance is the breaking of a wine glass when it is struck by a tuning fork.

| Type of oscillation | Description                                                                      | Amplitude           | Frequency                       | Force applied                          |
|---------------------|----------------------------------------------------------------------------------|---------------------|---------------------------------|----------------------------------------|
| Free oscillation    | Oscillates back and forth with a constant amplitude and frequency                | Constant            | Natural frequency of the system | Restoring force (Inertia)              |
| Forced oscillation  | Oscillates with a frequency that is equal to the frequency of the external force | Variable            | Frequency of the external force | Restoring force & External force       |
| Damped oscillation  | Loses energy over time and the amplitude decreases gradually                     | Decreases over time | Natural frequency of the system | Restoring force & Resistive force      |
| Resonance           | Amplitude is greatly increased                                                   | Increases           | Natural frequency of the system | Restoring force & large External force |

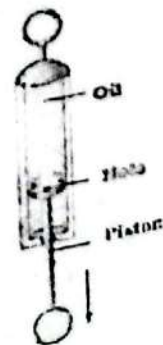
### PRACTICAL EXAMPLE OF DAMPED OSCILLATIONS

- Damping is the reduction in amplitude of an oscillating system due to the dissipation of energy.
- Shock absorber is a device that is used to dampen the oscillations of a spring.
- The shock absorber in a vehicle is a piston that is filled with viscous oil.
- When the vehicle encounters a bump, the piston compresses and the oil flows through the holes in the piston.
- The viscous force dissipates energy and the spring returns to its equilibrium length without oscillating up and down.

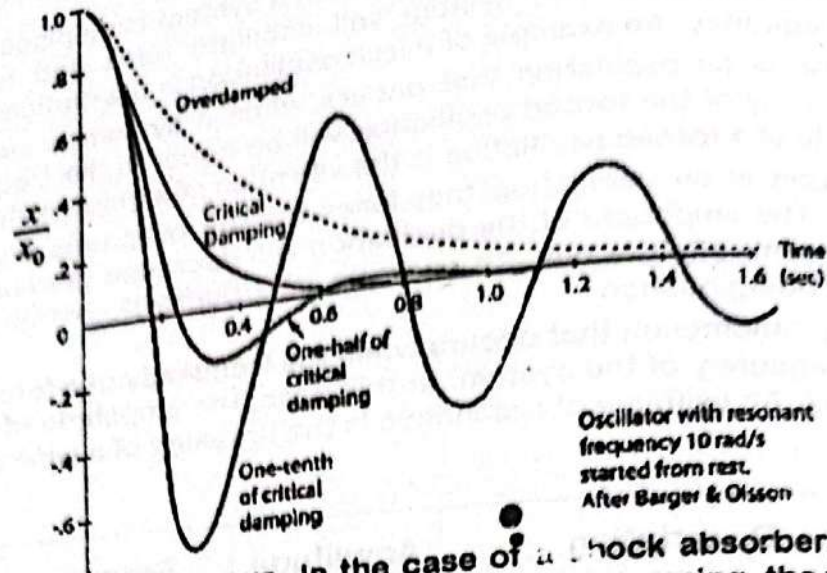
Shock absorber



Suspension, spring







Damping is not always disadvantageous. In the case of a shock absorber, it is actually beneficial because it helps to smooth out the ride of the vehicle. Without damping, the vehicle would bounce up and down excessively when it encounters bumps, which would be uncomfortable for the passengers.

### Frequency response and Sharpness of Resonance (Q-Factor):

- Oscillators are systems that vibrate or swing back and forth.
- Resonance is a phenomenon that occurs when the frequency of an external force matches the natural frequency of an oscillator.
- When resonance occurs, the amplitude of the oscillation increases.
- This can be harmful if the amplitude is too large, as it can lead to the collapse of structures.
- Damping is a force that opposes the motion of an oscillator.
- It can help to prevent resonance by reducing the amplitude of the oscillation.
- The Q factor is a measure of how sharp the resonance peak is.
- It is defined as the number of oscillations an oscillator makes before its amplitude decays to zero.
- A higher Q factor means that the resonance peak is sharper.
- The Q factor depends on the amount of damping in the system.
- A system with more damping will have a lower Q factor.

Q-Factor is a dimensionless quantity and it is given as,

$$Q = \frac{E_{\text{stored}}}{E_{\text{lost}}}$$



Multiple Choice Questions

MCQ'S

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ANS. |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <p>1. Two simple pendulums A and B with same lengths, and equal amplitude of vibrations, but the mass of A is twice the mass of B, their period are <math>T_A</math> and <math>T_B</math> and energies are <math>E_A</math> and <math>E_B</math> respectively. Choose the correct statement.</p> <p>a) <math>T_A = T_B</math> and <math>E_A &gt; E_B</math><br/> b) <math>T_A &lt; T_B</math> and <math>E_A &gt; E_B</math><br/> c) <math>T_A &gt; T_B</math> and <math>E_A &lt; E_B</math><br/> d) <math>T_A = T_B</math> and <math>E_A &lt; E_B</math></p> | a    |
| <p>2. In order to double the period of a simple pendulum:</p> <p>a) Its length should be doubled<br/> b) Its length should be quadrupled<br/> c) Its mass should be double<br/> d) Its mass should be quadrupled</p>                                                                                                                                                                                                                                                                                                                                         | b    |
| <p>3. A simple Harmonic oscillator has amplitude A and time period t. Its maximum speed is</p> <p>a) <math>4A/t</math><br/> b) <math>2A/t</math><br/> c) <math>4\pi A/t</math><br/> d) <math>2\pi A/t</math></p>                                                                                                                                                                                                                                                                                                                                             | d    |
| <p>4. A spring attached by a load of weight W is vibrating with a period T. If the spring is divided in four equal parts and the same load is suspended from one of these parts, the new time period is:</p> <p>a) <math>T/4</math><br/> b) <math>2T</math><br/> c) <math>T/2</math><br/> d) <math>4T</math></p>                                                                                                                                                                                                                                             | c    |
| <p>5. The total energy of a particle executing simple harmonic motion is proportional to:</p> <p>a) frequency of oscillation<br/> b) maximum velocity of motion<br/> c) amplitude of motion<br/> d) square of amplitude of motion</p>                                                                                                                                                                                                                                                                                                                        | d    |
| <p>6. A child swinging on a swing in sitting position, stands up, then the time period of the swing will:</p> <p>a) Increase<br/> b) decrease<br/> c) remains the same<br/> d) increases if the child is long and decreases if the child is short</p>                                                                                                                                                                                                                                                                                                        | b    |
| <p>7. If a body oscillates at the angular frequency of the driving force, then the oscillations are called:</p> <p>a) Forced oscillations<br/> b) Coupled oscillations<br/> c) Free oscillations<br/> d) Maintained oscillations</p>                                                                                                                                                                                                                                                                                                                         | a    |
| <p>8. A simple harmonic oscillator with a natural frequency <math>\omega_n</math> is forced to oscillate with a driving frequency <math>\omega_d</math>. The Resonance occurred when:</p> <p>a) <math>\omega_n &lt; \omega_d</math><br/> b) <math>\omega_n &gt; \omega_d</math><br/> c) <math>\omega_n = \omega_d</math><br/> d) <math>\omega_n \neq \omega_d</math></p>                                                                                                                                                                                     | c    |
| <p>9. In vehicles, shock absorber reduced the jerks:</p> <p>a) The shock absorber is the application of damped oscillations.<br/> b) Damping effect is due to the fractional loss of energy<br/> c) Shock absorbers in vehicles reduced jerk<br/> d) All of these</p>                                                                                                                                                                                                                                                                                        | d    |



10. A heavily damped system has a fairly flat resonance curve in:
- a) An acceleration time graph    b) An amplitude frequency graph  
c) Velocity time graph    d) Distance-time graph

b

## \*\*\*\*\* Exams Practice MCQs \*\*\*\*\*

|                                                                                                                                                                                            |                                                                                                                 |   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---|
| 1. If the mass of the bob of a simple pendulum is doubled, its time period will                                                                                                            | (a) doubled<br>(b) Halved<br>(c) remain constant<br>(d) quadrupled                                              | c |
| 2. The frequency of a second's pendulum is:                                                                                                                                                | (a) 1 Hz<br>(b) 2 Hz<br>(c) 0.5 Hz<br>(d) 4 Hz                                                                  | a |
| 3. The value of elastic restoring force in case of a spring is:                                                                                                                            | (a) $Kx$<br>(b) $-Kx$<br>(c) $\frac{1}{2}Kx$<br>(d) $-\frac{1}{2}Kx$                                            | a |
| 4. If the bob of a simple pendulum is replaced by another bob of double mass but of the same size, its time period:                                                                        | (a) Increases<br>(b) Decreases<br>(c) Remains the same<br>(d) becomes infinity                                  | c |
| 5. The time period of a simple pendulum depends upon its:                                                                                                                                  | (a) Length<br>(b) Amplitude<br>(c) Temperature<br>(d) Mass of the bob                                           | a |
| 6. An object is executing simple harmonic motion. Its kinetic energy is maximum at its:                                                                                                    | (a) mean position<br>(b) extreme position<br>(c) at any point along the path<br>(d) None of these               | b |
| 7. A spring mass system is performing simple harmonic motion will the time period 'T'. If we double the mass of its bob, the new time period will be:                                      | (a) T<br>(b) 2T<br>(c) 1.414T<br>(d) 0.707 T                                                                    | d |
| 8. The acceleration of free fall on the moon is $\frac{1}{6}$ of that on the earth. What would be the period on the moon of a simple pendulum which has a period of 1 second on the earth. | (a) $\frac{1}{6}$ sec.<br>(b) $\frac{1}{6}$ sec<br>(c) 6 sec<br>(d) 12 sec                                      | c |
| 9. The frequency of oscillation of a simple pendulum depends upon:                                                                                                                         | (a) The mass of the bob<br>(b) The amplitude of vibration<br>(c) The length of the pendulum<br>(d) None of them | a |
| 10. What is the defining characteristic of damped oscillations?                                                                                                                            | (a) Constant amplitude<br>(b) Increasing amplitude<br>(c) Decreasing amplitude<br>(d) No amplitude              | c |
| 11. In a damped oscillatory system, the damping force is proportional to:                                                                                                                  | (a) Displacement<br>(b) Velocity<br>(c) Acceleration<br>(d) Mass                                                | b |
| 12. What happens to the amplitude of a free oscillation over time?                                                                                                                         | (a) It remains constant                                                                                         | b |



- (b) It decreases  
(c) It increases  
(d) It varies unpredictably

13. Which of the following is a common source of damping in mechanical oscillatory systems?

- (a) Friction  
(b) Gravity  
(c) Tension  
(d) Elasticity

14. In a forced oscillation, the driving frequency matches the natural frequency of the system. What is the term for this condition?

- (a) Resonance  
(b) Damping  
(c) Harmonics  
(d) Periodicity

15. In a damped forced oscillation, what is the phase relationship between the driving force and the displacement of the system at resonance?

- (a) 0 degrees (in-phase)  
(b) 90 degrees (out of phase)  
(c) 180 degrees (opposite phase)  
(d) It varies with damping.

16. In a simple harmonic oscillator, when the displacement from the mean position is maximum, which energy is maximum?

- (a) Kinetic Energy  
(b) Potential Energy  
(c) They are equal  
(d) It depends on the specific oscillator

17. At the mean position of a simple harmonic oscillator, the kinetic energy is:

- (a) Zero  
(b) Maximum  
(c) Equal to potential energy  
(d) Half of the potential energy

18. In an oscillatory system, which type of energy is associated with the displacement of the object from its equilibrium position?

- (a) Kinetic Energy  
(b) Potential Energy  
(c) Both Kinetic and Potential Energy  
(d) None of the above

19. The total mechanical energy of an oscillator in simple harmonic motion:

- (a) Increases with time  
(b) Decreases with time  
(c) Remains constant  
(d) Depends on the mass of the oscillator

20. In a simple harmonic oscillator, which energy is at its maximum when the displacement is zero?

- (a) Kinetic Energy  
(b) Potential Energy



- (c) Both are zero
- (d) They are always equal

21. A particle is executing SHM with amplitude  $A$  and angular frequency  $\omega$ . What is the phase angle of the particle at a time  $t = T/4$ ?

- (a)  $0^\circ$
- (b)  $90^\circ$
- (c)  $180^\circ$
- (d)  $270^\circ$

## Chapter Formulae and Numericals

|                                                 |                                                 |
|-------------------------------------------------|-------------------------------------------------|
|                                                 | $F = -kx$                                       |
| Hooke's Law                                     | $a \propto -x$                                  |
| Simple Harmonics Motion (S.H.M)                 | $x = x_0 \cos \theta$                           |
| Relation B/w Uniform Circular Motion and S. H.M | $x = x_0 \cos \omega t$                         |
| Displacement of Projection Q                    | $V_Q = x_0 \omega \sqrt{1 - \frac{x^2}{x_0^2}}$ |
| Velocity of Projection Q                        |                                                 |
| Max. Velocity of Projection Q                   | $V_{\max} = x_0 \omega$                         |
| For Oscillating Mass attached to a string       | $\omega = \sqrt{\frac{k}{m}}$                   |
| Angular Velocity                                |                                                 |
| Phase                                           | $x = x_0 \cos(\omega t + \varphi)$              |
| Time Period of Spring Mass Oscillator           | $T = 2\pi \sqrt{\frac{m}{k}}$                   |
| Time Period of Simple Pendulum                  | $T = 2\pi \sqrt{\frac{l}{g}}$                   |
| Angular velocity of simple pendulum             | $\omega = \sqrt{\frac{g}{l}}$                   |
| Frequency                                       | $f = \frac{1}{T}$                               |
| K.E of Oscillator                               | $K.E = \frac{1}{2} k(x_0^2 - x^2)$              |
| P.E of Oscillator                               |                                                 |
| Total Energy of Oscillator                      | $PE = U = \frac{1}{2} kx^2$                     |
| Q - Factor                                      | $E = \frac{1}{2} kx_0^2$                        |
|                                                 | $Q = \frac{E_{\text{stored}}}{E_{\text{lost}}}$ |